

IsaacSim及IsaacLab安装（Linux）

安装需求

系统需求

Element	Minimum Spec	Good	Ideal
OS	Ubuntu 20.04/22.04 Windows 10/11	Ubuntu 20.04/22.04 Windows 10/11	Ubuntu 20.04/22.04 Windows 10/11
CPU	Intel Core i7 (7th Generation) AMD Ryzen 5	Intel Core i7 (9th Generation) AMD Ryzen 7	Intel Core i9, X-series or higher AMD Ryzen 9, Threadripper or higher
Cores	4	8	16
RAM	32GB [1]	64GB [1]	64GB [1]
Storage	50GB SSD	500GB SSD	1TB NVMe SSD
GPU	GeForce RTX 3070	GeForce RTX 4080	RTX Ada 6000
VRAM	8GB [1] [2]	16GB	48GB

驱动需求

Driver Version Support	Windows	Linux
Recommended	537.58 (GameReady, Studio), 537.70 (RTX/Quadro, Grid/vGPU)	535.129.03 (GameReady, Studio, RTX/Quadro, Grid/vGPU)
Minimum	537.58 (GameReady, Studio), 537.70 (RTX/Quadro, Grid/vGPU)	535.129.03 (GameReady, Studio, RTX/Quadro, Grid/vGPU)

IsaacSim安装

本体安装

1. 下载安装包

2. 运行命令 `./post_install.sh`

3. 运行命令 `./isaac-sim.selector.sh`，直接点击start后出现界面即为成功

资源包安装

1. 下载三个部分的资源包

2. 解压安装包（注意要将三个包合并到同一个根文件夹中）：

代码块

```
1  mkdir ~/isaacsim_assets
2  cd ~/Downloads
3  unzip "isaac-sim-assets-1-4.5.0.zip" -d ~/isaacsim_assets
4  unzip "isaac-sim-assets-2-4.5.0.zip" -d ~/isaacsim_assets
5  unzip "isaac-sim-assets-3-4.5.0.zip" -d ~/isaacsim_assets
```

3. 设置资源包默认路径，编辑isaacsim安装目录下 `apps/isaacsim.exp.base.kit` 文件，底部添加：（记得替换成自己的目录）

代码块

```
1  [settings]
2  persistent.isaac.asset_root.default="/home/cqy/isaacsim_assets/Assets/Isaac/4.5"
3
4  exts."isaacsim.asset.browser".folders = [
5      "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Robots",
6      "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/People",
7      "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/IsaacLab",
8      "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Props",
9      "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Environments",
10     "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Materials",
11     "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Samples",
12     "/home/cqy/isaacsim_assets/Assets/Isaac/4.5/Isaac/Sensors",
13 ]
```

4. 启动isaacsim，并点击isaacsim assets browser中齿轮图标检查默认资产根路径

代码块

```
1  ./isaac-sim.sh -
   -/persistent/isaac/asset_root/default="/home/cqy/isaacsim_assets/Assets/Isaac/4.5"
```

IsaacLab安装

1. 将isaasim添加到环境变量（isaacsim路径要和真实安装路径一致）：

代码块

```
1  # Isaac Sim root directory
2  export ISAACSIM_PATH="${HOME}/isaacsim"
3  # Isaac Sim python executable
4  export ISAACSIM_PYTHON_EXE="${ISAACSIM_PATH}/python.sh"
```

2. 克隆IsaacLab: `git clone git@github.com:isaac-sim/IsaacLab.git`
3. cd到IsaacLab目录下创建符号链接: `ln -s ${ISAACSIM_PATH} _isaac_sim`
4. 创建conda虚拟环境: `./isaacLab.sh --conda isaacLab`
5. 安装依赖: `sudo apt install cmake build-essential`
6. 安装扩展: `./isaacLab.sh --install`
7. 验证安装:

代码块

```
1  python scripts/tutorials/00_sim/create_empty.py
2  或者
3  python scripts/reinforcement_learning/rsl_rl/train.py --task=Isaac-Ant-v0 --headless
```

8. 设置vscode: 打开vscode设置, 选择run task, 选择setup_python_env

在vscode中找到setting.json, 在 `python.analysis.extraPaths` 最后加上isaacLab库的路径, 例（注意换成自己的isaacLab路径）：

代码块

```
1  "/home/cqy/opensource_projects/IsaacLab/source/isaacLab",
2  "/home/cqy/opensource_projects/IsaacLab/source/isaacLab_tasks",
3  "/home/cqy/opensource_projects/IsaacLab/source/isaacLab_assets",
4  "/home/cqy/opensource_projects/IsaacLab/source/isaacLab_rl"
```